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A Deep Learning Model for Automated Classification of Intraoperative Continuous EMG

Xuefan Zha,

Carnegie Mellon University, Pittsburgh, PA, USA

Leila Wehbe,

Carnegie Mellon University, Pittsburgh, PA, USA

Robert J. Scwabassi [Senior Member, IEEE],

Computational Diagnostics, Inc., Pittsburgh, PA, USA; Neuroscience Institute, Allegheny Health Network, Pittsburgh, PA. USA..

Zachary Mace,

Computational Diagnostics, Inc., Pittsburgh, PA, USA; Neuroscience Institute, Allegheny Health Network, Pittsburgh, PA. USA..

Ye V. Liang,

Computational Diagnostics, Inc., Pittsburgh, PA, USA; Neuroscience Institute, Allegheny Health Network, Pittsburgh, PA. USA..

Alexander Yu,

Allegheny Health Network, Pittsburgh, PA, USA.

Jody Leonardo,

Allegheny Health Network, Pittsburgh, PA, USA.

Boyle C. Cheng,

Allegheny Health Network, Pittsburgh, PA, USA.

Todd A. Hillman,

Pittsburgh Ear Associates, Pittsburgh, PA, USA.

Douglas A. Chen,

Pittsburgh Ear Associates, Pittsburgh, PA, USA.

Cameron N. Riviere* [Senior Member, IEEE]

Carnegie Mellon University, Pittsburgh, PA, USA

Abstract

Objective: Intraoperative neurophysiological monitoring (IONM) is the use of electrophysiological methods during certain high-risk surgeries to assess the functional integrity of nerves in real time and alert the surgeon to prevent damage. However, the efficiency of IONM in

* camr@ri.cmu.edu.

current practice is limited by latency of verbal communications, inter-rater variability, and the subjective manner in which electrophysiological signals are described.

Methods: In an attempt to address these shortcomings, we investigate automated classification of free-running electromyogram (EMG) waveforms during IONM. We propose a hybrid model with a convolutional neural network (CNN) component and a long short-term memory (LSTM) component to better capture complicated EMG patterns under conditions of both electrical noise and movement artifacts. Moreover, a preprocessing pipeline based on data normalization is used to handle classification of data from multiple subjects. To investigate model robustness, we also analyze models under different methods for processing of artifacts.

Results: Compared with several benchmark modeling methods, CNN-LSTM performs best in classification, achieving accuracy of 89.54% and sensitivity of 94.23% in cross-patient evaluation.

Conclusion: The CNN-LSTM model shows promise for automated classification of continuous EMG in IONM.

Significance: This technique has potential to improve surgical safety by reducing cognitive load and inter-rater variability.

Keywords

Electromyography; Intraoperative Neuromonitoring; Convolutional neural networks; Pattern recognition

I. Introduction

INTRAOPERATIVE neurophysiological monitoring (IONM) is used during certain high-risk operations to assess the effects of surgical manipulations on the structure–function relationships of the patient’s nervous system (NS). IONM closes an information loop around the surgeon, patient, and anesthesiologist to alert the surgical team to the possibility of impending injury and thus to prevent permanent damage to the patient’s nervous system. Neurophysiological data are obtained continuously throughout the procedure and interpreted in real time by a physician with specialty training in clinical neurophysiology. Any remarkable changes are reported to the surgical team during the operation [1]. There are two main problems in the process of IONM. Firstly, during surgical procedures, abnormal changes in observed signals may occur rapidly and there may be an unacceptable latency in communication between the monitoring team and the surgeon which can exacerbate injury. Secondly, different levels of training and experience may produce differences in the recognition of abnormal neural activity, resulting in either delayed or inadequate adjustments of operative technique, leading to avoidable injury; i.e., in current practice, both human interpretation and human use of the IONM information are qualitative and subjective [2]. To address these problems, in this preliminary study, we have investigated the use of an automatic signal classification system for abnormal condition detection during IONM. The long-term goal of our effort is to include such monitoring within the control system of a surgical robotic system as an intraoperative safety check [3].

Electromyogram (EMG) monitoring is a useful indicator of the integrity of descending activity in the spinal cord, particularly in the peripheral nerves and their interface to various muscles, which is commonly used to provide accurate warning signs during surgery [4]. EMG activity consists of motor unit action potentials whose characteristics provide information about the relationships between particular nerves and muscles. EMG is used also in the diagnosis and assessment of neuromuscular disorders [5].

For the traditional EMG pattern recognition task, several steps are involved: firstly, EMG is preprocessed to remove noise, then a time window is used to segment the data to account for its nonstationary properties, then feature extraction techniques are utilized. Due to the great variation in EMG signals, many EMG featurizations have been proposed [6]. There are still no general rules for determining which feature is best to use in different tasks. The final step is to use data features in a classifier or regressor to categorize the EMG pattern [7]. High classification accuracy has been achieved in several previous studies. Lucas et al. used the discrete wavelet transform of the EMG signal with unconstrained parameterization and a multi-class nonlinear support vector machine for myoelectric prosthesis control [8]. In recent years, numerous machine learning algorithms have been applied to EMG pattern detection either for the assessment of neuromuscular pathology or for gesture recognition in robot control. Neural networks have also been used widely in EMG classification, for example deep belief networks which have performed well at capturing nonlinear and time-varying characteristics [9]. Xia et al. designed a recurrent neural network (RNN) to estimate limb movement based on measurements from healthy subjects [10]. Zhai et al. designed a convolutional neural network (CNN) to operate on NinaPro Datasets 2 and 3 [11] for neuroprosthesis control task and achieved accuracy of about 83% [12]. Xie et al. [13] compared the performance of different classifiers and reported that a hybrid model containing one convolutional layer and one recurrent layer presented the best performance. Hu et al. [14] also proposed an attention-based hybrid CNN-RNN architecture for sEMG-based gesture recognition, which presented better performance compared to the CNN module of the proposed architecture on a different database. It has been demonstrated that this models both the spatial and temporal information of sEMG and focuses on the subsegments which contain more discriminative information for gesture recognition.

In most traditional machine learning methods for EMG classification tasks, EMG signal features such as mean absolute value, root mean square, and average amplitude change are extracted before classification [15]. However, in IONM EMG signals, it is difficult to analyze feature information to detect sEMG abnormal pattern due to the complicated nature of the intraoperative waveforms. This suggests the need for methods which combine time-domain and frequency-domain information. Previous research has shown that for complicated EMG signal classification tasks, neural networks have an advantage over other state-of-the-art machine learning methods [16]. These previous results have motivated us to apply neural networks to the IONM EMG classification task.

According to Romstck et al., EMG activity can be divided into five typical EMG waveform patterns, which they refer to as spikes, bursts, and A, B, and C trains [4]. (It should be noted that this and subsequent studies [17][18] dealt with the facial nerve, whereas data for the present study were obtained from the recurrent laryngeal nerve.) In their study, A trains,

which are a distinct EMG waveform of sinusoidal pattern that produces a high-frequency sound have been shown to have strong relationship with long-term nerve injury. B trains are similar to myokymic discharges, but the pathophysiological mechanism underlying this pattern is not clear. C trains are similar to an interference pattern during maximum voluntary muscle contraction, although patients are under deep steady-state anesthesia. Spikes and bursts are either elicited by direct nerve contact with a surgical instrument or arise spontaneously at any time during surgery [4]. Thus we are concerned more with train patterns, which are related to the actual nerve condition.

Generally, the amplitude range of the subdermal EMG (sEMG) signal is 0-10 mV peak to peak prior to amplification. There is contamination in sEMG signals due to noise in electronics equipment, ambient noise, and motion artifacts[19]. In order to reduce these effects, general filtering and correction methods are used in EMG preprocessing, such as signal correction (compensating unexpected sudden change from baseline), high pass filters, and median filters[20]. However, in the IONM setting, these techniques may not work since meaningful patterns may also be removed using filter and correction methods. As an example, Fig.1 demonstrates the loss of spike activity due to typical preprocessing.

This paper presents the first investigation into deep learning methods for automatic recognition of EMG patterns during IONM. We report on a hybrid convolutional neural network - recurrent neural network (CNN-RNN) model to automatically classify abnormal subdermal EMG signals obtained during surgery. Data from thyroid surgery was utilized since this provides an experimental model in which a single muscle is innervated by a single nerve. Our approach includes several innovations such as a simplified preprocessing pipeline that minimizes the loss of information from the signal and handles classification of data from multiple subjects. We compare the performance of our proposed model with several baselines. Our approach exceeds in performance all baselines and can effectively detect abnormal EMG patterns under conditions of both noise and artifacts.

II. METHODS

The data we used are de-identified single-channel continuous EMG obtained from the recurrent laryngeal nerve during thyroid surgery from five patients under a board-approved data use agreement. All data were acquired using a standard methodology consistent with international guidelines [21][22]. EMG data were recorded continuously using a Dragonfly electrode (LSE500, Neurovision Medical Products, Ventura, CA) placed on standard endotracheal tubes, appropriately sized. A NeuroNet 600 system (Computational Diagnostics, Inc, Pittsburgh, PA) was used to obtain the data. Data were sampled at 12 kHz, with a bandpass filter with a passband from 1 to 4000 Hz, and downsampled to 1200 Hz. There was no drilling during the surgical exposure and all artifacts observed were secondary to electrosurgical manipulations. Recorded data, though continuous, were organized into epochs of 1 second for classification.

A. Expert Annotation of Data

Expert neurophysiologists classified the EMG data epoch by epoch into five main types which are shown in Table I: Injury, Burst, Irritation, Evoked EMG, and Quiet. The Injury

pattern is similar to the A trains previously mentioned. Injury activity is seen to have a more or less rhythmic sequence. A burst is an isolated complex of superimposed spikes having spindlelike shapes, whose onset is abrupt and usually preceded by isolated spikes of low amplitude. Burst indicates working near a nerve and irritating the nerve in a strong fashion. Irritation is characterized by continuous irregular EMG activity that is composed of numerous overlapping components, which provide an alert of possible impending injury to the nerve. Irritation amplitude and frequency allow it to be divided into four categories: baseline, weak, regular, and strong. Evoked EMG refers to EMG activity produced by electrical stimulation of the nerve used to assess integrity of the nerve; evoked EMG and has a stereotyped shape of a singular wave which tends to be triphasic in shape, similar to a compound muscle potential. The last sEMG type is Quiet, which indicates the usual healthy EMG baseline activity; this represents the "hoped-for" EMG activity during an operation. The data also contain artifacts which are related to electrode manipulation and static from instruments. Fig.2 shows different EMG signal types. Fig.3 shows detail of the several grades of irritation in the data.

Injury and Burst rarely occur in patient data, since irritation activity usually alerts the operative team to proximity to a nerve, which they then seek to avoid. Evoked activity is inherently easier to identify, since the timing of applied stimuli is known. Therefore we focus primarily on recognition of irritation, which serves as a sign of the imminent possibility of injury.

B. Normalization of Data

To avoid the problem of general EMG preprocessing and to combine time- and frequency-domain analysis, the Short-Time Fourier Transform (STFT) is used to normalize the sEMG signal. The STFT is one of the most widely used methods to extract frequency information within nonstationary signals. Each coefficient of the STFT represents the magnitude of a frequency sub-band which, in our case, contains information related to muscle EMG[23].

Firstly, we apply STFT with a sliding Hanning window with a length of 100 samples and 80% overlap between windows to obtain the frequency-band power-density distribution for every sliding window. Then we use the mean and standard derivation of power density for every frequency band in 'quiet' epochs to normalize frequency band power density of all subsequent sliding windows (the 'quiet' epochs are recorded before surgical intervention begins) as below:

$$X_{normalized} = \frac{(X - \bar{X}_{quiet})}{\sigma_{quiet}}, \quad (1)$$

where $X_{normalized}$ is normalized signal spectrogram and X is raw signal spectrogram after STFT, $\bar{X}_{quiet}$ is mean of quiet epoch spectrograms and σ_{quiet} is standard derivation of quiet epoch spectrograms. Example spectrograms for normalized irritation and normal signal are shown in Fig.4.

To make our model work both on a single patient and across patients, different preprocessing methods are used. For the single-patient model, data normalization uses only one patient's

‘quiet’ epochs as reference. For the cross-patient model, 100 random ‘quiet’ epochs were selected from different patients during the model training with this reference reused in inference time.

Note that avoiding the usual filtering and preprocessing methods and computing the normalized spectrograms allows us to preserve information in all frequency bands. The normalization step does not remove information but simply rescales it to highlight deviations from the benchmark state. Because of this, the algorithm is able to focus on aspects of the signal that are most useful for classification. For example, large amplitude slow drifts are typically not used by neurophysiologists for labeling a signal. Neurophysiologists might focus instead at changes at higher frequencies that are of smaller amplitude. Our normalized spectrograms preserve all the information and allows the classifier to learn the informative features and ignore the less useful ones.

C. CNN-LSTM Model

Irritation is distinguished from non-irritation by the presence of vertical stripes of high amplitudes at mid-to-high frequencies. The onset of these high-amplitude bands is not locked to the onset of the epoch, and we therefore need a flexible model that can detect these patterns irrespective of when in an epoch they occur[24]. To capture these patterns, we built a hybrid model with a convolutional Neural Network (CNN) component and a Long Short-Term Memory (LSTM) component.

The convolutional neural network (CNN) is the most widely used deep learning technique for sEMG-based recognition[16]. CNNs works well for identifying simple patterns which are then used to form more complex patterns within higher layers[25].

The RNN contains feedback loops and encodes contextual information of a temporal sequence. As standard RNN suffers from a gradient-vanishing or exploding problem, LSTM has been proposed to alleviate this issue[26].

Our CNN-LSTM model has two main parts: CNN and LSTM, as shown in Fig.5. The CNN component acts as a feature extractor to transform a normalized sequence into spatiotemporal feature vectors. To deal with the problem of reducing the amount by which the hidden unit values shift around, batch normalization is added after convolutional layers. In order to extract feature vectors in a frequency band, the CNN part is composed of two convolution blocks. Each block has a kernel size of (3,1) with stride of 1 and one batch normalization layer. The LSTM component acts as a predictor that combines the current CNN output with the output from previous epochs to predict the irritation state. Here we use a two-layer bidirectional LSTM with 256 hidden units to extract the time relationship between feature sequences. The proposed architecture models the spatiotemporal information of the input EMG signals and the layered configuration of the proposed network as shown in Table II. After data normalization, a spectrogram feature matrix for the EMG signal is constructed and put into the model. Data are stacked epoch by epoch.

D. Baseline Models

We compare the performance of the CNN-LSTM model to multiple baselines to verify that the observed performance is a function of the architecture used and that a simpler architecture would not perform as well. We use as baselines a CNN network alone and an LSTM network alone, as well as Gaussian Naive Bayes (GNB) [27].

GNB is a statistical classification technique based on Bayes' Theorem. It is one of the simplest supervised learning algorithms. GNB assumes input features are normally distributed and independent of the other features given the label. Predictions are then computed using:

$$P(Y | X) = P(Y) \frac{P(X | Y)}{P(X)} \propto P(Y) \prod_{j=1}^m P(X^j | Y), \quad (2)$$

in which Y is the label of data point X with X having m dimensions indexed by j .

III. Experiment

A. Experiment Setting

Before applying our model on test data, we did STFT window-size analysis for data normalization. In order to find the best STFT window size in the data normalization step to detect irritation patterns, we compared different sizes of sliding windows to extract a spectrogram feature matrix. Results were evaluated based on sensitivity in detection of irritation for the validation data.

For the single-patient model, we chose three patients whose data contained all EMG signal classes (the other two did not). We compared our architecture with CNN and LSTM separately. We randomly chose 85% for training and 15% for testing within one patient's data. To compare training of different models, CNN-LSTM was compared with CNN, LSTM and Nave Bayes separately. Nave Bayes was used as the baseline of a simple model. CNN means that we only used the CNN part of the model as a training model, which we called two-layer CNN. RNN means that we only used the LSTM part of our model, which we called two-layer RNN. All four models were trained on the same training dataset and testing dataset separately for three individual patients.

In order to transfer training results between patients, we used CNN-LSTM and analyzed its performance in the cross-patient model; we used four patients for training and one for testing. The details regarding training and testing dataset size are shown in Table III. We also used the identification of artifacts in data in order to compare cross-patient model robustness under the influence of artifacts. In this step we compared two types of data: raw data including the electric artifacts, and data from which the epochs containing electric artifacts had been removed.

To illustrate the necessity of data normalization, we also compared training with data normalization and without data normalization in the single-patient model and the cross-patient model separately.

Each deep learning model is trained with 45 epochs with the Adam optimizer[28], with a learning rate of 0.001 and batch size of 100.

B. Indexing Method

A twofold indexing method was used, based on neurophysiologists' labeling method. We built an index for training that matched the human labeling, including 7 classes. In order to simplify our problem and focus more on clinically significant signals, we also built a simplified index for training. In the simplified index, regular irritation, strong irritation, baseline irritation, and weak irritation are all indexed as irritation and other classes are indexed as quiet. The details of each class are shown in Table III. To provide clinically significant prediction, we also compared the original index method and simplified index method in the cross-patient model.

C. Evaluation

Our results were evaluated in terms of average accuracy on the test set. We also reported the sensitivity of the irritation classification (proportion of correctly identified irritation samples out of all irritation samples) since detecting nerve irritation is one of the most important goals during surgery. We computed confidence intervals (CI) for each measure as a function of the error e :

$$CI = e \pm z_N \sqrt{\frac{e(1-e)}{N}} \quad (3)$$

where N is the total number of samples. We use $z_N = 1.96$ to obtain a 95% confidence interval [29].

IV. RESULTS

A. STFT Window Size Analysis

Both the cross-patient model and the single-patient model were evaluated in terms of sensitivity in detecting irritation under different window sizes on validation data chosen as a subset of the original training data. As Table IV shows, a Hanning window of length 100 is best for our model for both single-patient training and cross-patient training.

B. Single-patient Model Analysis For Different Models

Different model test results from three patients separately are shown in Table V. We used three models with different complexity as a baseline. On all three patients, the GNB model performs much worse than the CNN-LSTM model. The simple design of GNB is not powerful enough to deal with the classification task on different patients. The deep learning models have noticeably different performances on different patients. For example, the RNN model has very varied performance on different patients. It has very low accuracy and sensitivity for patient 1, much lower than GNB, while performing well for patients 2 and 3. It appears that different data distributions across patients adversely impact the performance and generalizability of the RNN model. The CNN model is more consistent across patients, indicating that a lot of information about the label can be found in the spectrogram of the

current epoch. However, the CNN model does not perform as well as the CNN-LSTM model, which consistently provides stable performance (relatively tight confidence interval with higher accuracy and sensitivity) on different patients' data. This indicates that combining information across time is also useful for the classification task.

C. Cross-patient Model Analysis For Different Index Methods

Training and testing results for cross-patient models are shown in Table VI. Here we can see that the CNN-LSTM can successfully transfer training results between different patients. Before using the cross-patient model, we perform cross-patient normalization. From the table we can see that the simplified index method has better sensitivity with a tighter confidence interval for irritation prediction, which is expected since the classification has been simplified. However, the normal index method can provide detailed class information and also achieve good performance. The orange line and blue line in Fig. 6 show loss curves for training for the normal index and simplified index respectively. From the figure we can see the model trained to classify the simplified index has better initialization and converges faster than the normal index.

D. Normalization Analysis For Single Patient and Cross-Patient Classification

Figure 6 shows the loss curves of training on the normalized data (orange line) and data without normalization (pink line) for the cross-patient model and Fig.7 shows the loss curves of normalized data (red line) and data without normalization (blue line) for the single-patient model. We use a binary crossentropy loss (decrease in the loss corresponds to increase in accuracy). From these figures we can see that the model converges in around 20 iterations and that adding data normalization makes the single-patient training process converge even more quickly. However, in the cross-patient model data normalization matters even more. Without data normalization, training does not converge at all and is stuck at accuracy of 68%, due to the difference in the data distribution across subjects. When we do cross-patient data normalization, we use a more generalized reference to normalized data and without that the model does not transfer successfully between patients.

E. Robustness of Detection of Irritation

Prediction results for both raw data and artifact-free data are shown in Table VII. The loss curves for the training are shown in Fig. 8. The loss curves are similar for the two cases, thus removal of artifacts does not influence the model greatly. The CNN-LSTM cross-patient model is robust in the presence of artifacts.

F. Analysis of Misclassification

Two misclassification examples from patient 1 are shown in Fig. 9. Epoch 3917 was labeled as weak irritation but classified as quiet. Epoch 979 was labeled as artifact but classified as irritation. Inconsistent judgements made by people and classifiers are more likely to happen in transitional periods between one class and another. Sometimes continuous artifact also causes misclassification, because it seems like an irritation pattern.

V. Discussion

We have evaluated the CNN-LSTM model with data normalization in both single-patient and cross-patient conditions. The main finding is that the proposed model has robust performance under the influence of noise and artifact in various experimental settings.

The result of the single-patient analysis demonstrates that CNN-LSTM has higher accuracy in irritation classification in different patient data compared to several baseline models. From the experimental result we can see that a simpler machine learning method like GNB has lower performance on different patients, which suggests that the classification task requires a more complex model to recognize the irritation pattern because of the highly overlapping feature distribution. The CNN and RNN models can capture EMG features to some extent, but fail to have good classification performance in all patients. CNN-LSTM uses the CNN component to capture frequency-band features and uses the LSTM component to capture temporal structure in those features, effectively combining the advantages of CNN and RNN.

Since our aim is to transfer classification results between patients, in the cross-patient analysis we also focus on how to improve the performance of the CNN-LSTM model. In our experiment with different index methods, we found that the simpler index classification simplified the problem and succeeded in enabling the model to classify more accurately. In our experiment with artifacts, the model has good performance both with and without artifacts. This demonstrates that the CNN-LSTM model is robust under the influence of artifact.

In the data normalization analysis in single-patient and cross-patient models, normalization plays an important role in model training; this is especially so in the cross-patient models. Using data normalization to replace generic preprocessing avoids loss of information. An important feature in the cross-patient normalization is the fact that the normalization merges baseline EMG signal information from multiple patients.

This paper has reported on the application of the machinelearning-based EMG classification technique to a limited number of patients. For simplicity, the technique has been applied here to single-channel EMG data from thyroid surgery. Future work will require testing of the approach with data from a larger number of patients, and from procedures other than thyroid surgery, including procedures that involve multiple channels of IONM. It is noted that, once trained, the existing algorithm is suitable for implementation in real time. Implementation in clinical systems would therefore be straightforward, if performance were validated successfully with a suitably large dataset.

VI. Conclusion

This study demonstrates that deep learning is capable of automatically classifying continuous EMG patterns typically seen in IONM. The results of data normalization indicate that a simple preprocessing method can also effectively help a neural network to detect important EMG pattern under noise and artifact influence. Automatic irritation pattern

recognition has potential significance for provision of real-time alarming in IONM to assist neurophysiologists and provide additional safety for the patient.

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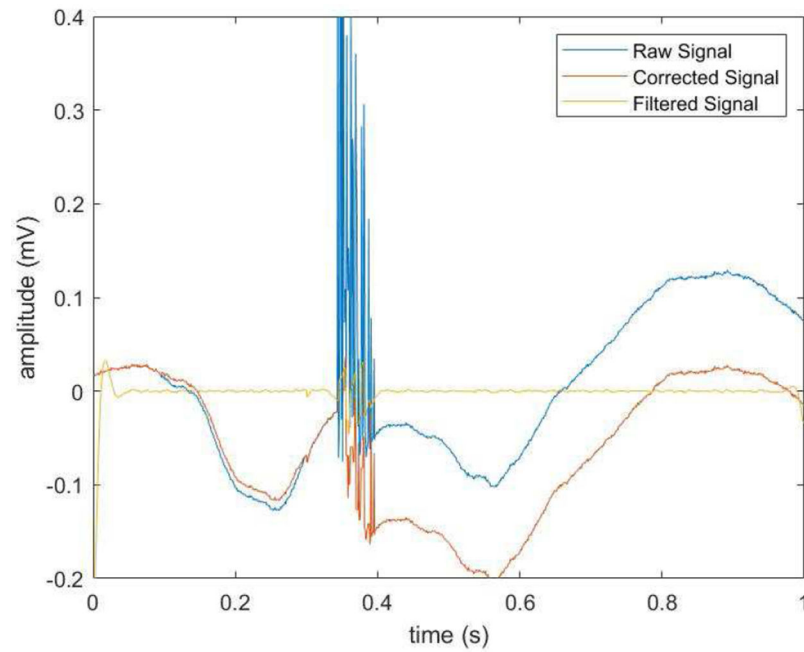


Fig. 1:

Illustration of problems that may be caused by preprocessing: raw EMG signal, corrected EMG signal, and signal processed by high pass filter and median filter of epoch 983 for patient 3. Epoch 983 is an example of artifact, but after correction and filtering, its waveform resembles a burst pattern, as shown in Table I. These preprocessing operations thus may introduce error into the classification process.

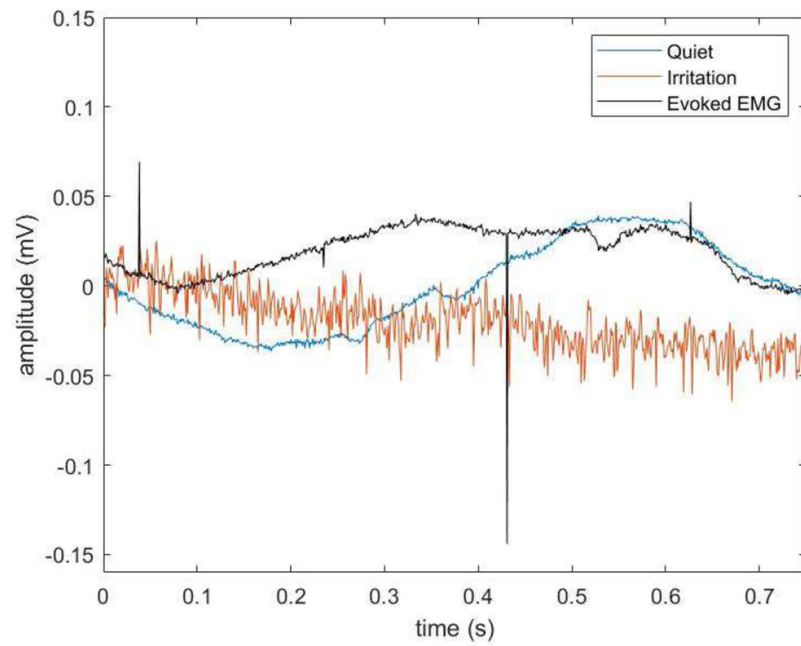


Fig. 2:
Common EMG patterns in IONM. Blue trace: Quiet data, demonstrating negligible activity. Brown trace: Irritation activity demonstrating high frequency components characteristic of this state. Black trace: Evoked compound muscle response demonstrating a compound muscle action potential as seen on a continuous data stream. Sharp peaks are electrical artifacts.

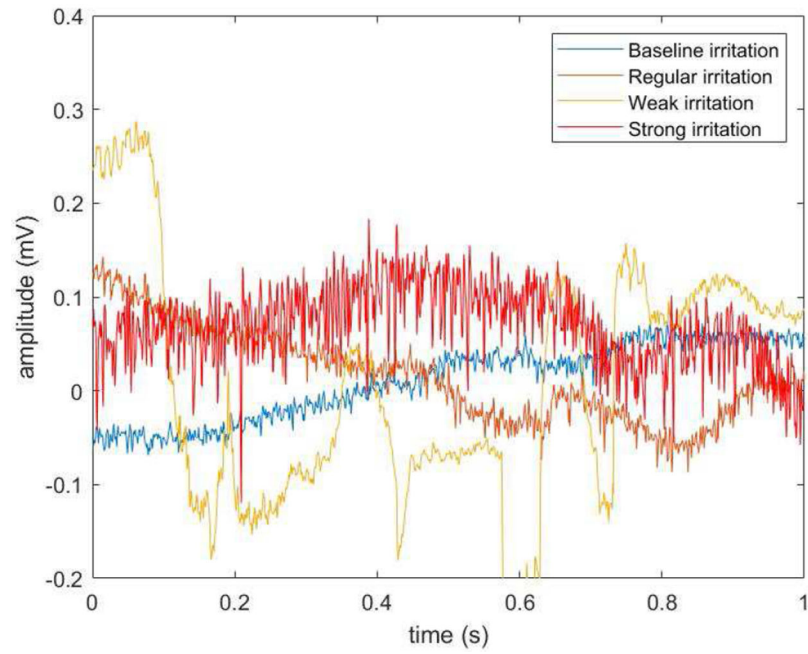


Fig. 3:

EMG Irritation patterns seen in IONM. Baseline irritation represents a relatively small irritation pattern which indicates slight stress on the nerve. Regular irritation is a typical irritation pattern with standard derivation within $10\mu V$. Strong irritation is an irritation pattern with relatively large amplitude and standard derivation above the baseline. Weak irritation is intermittent, with small amplitude.

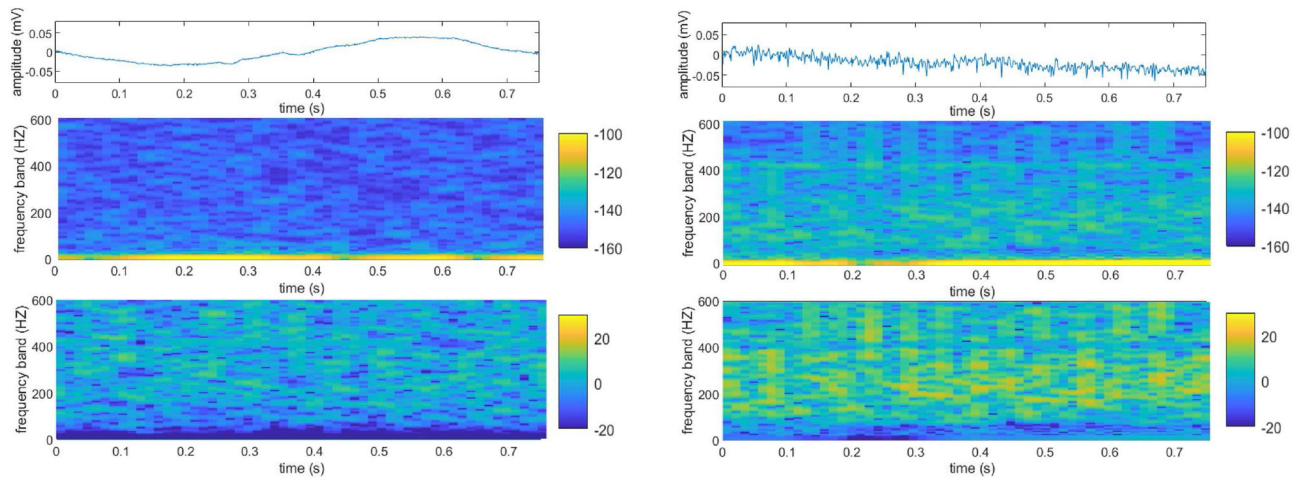


Fig. 4:

Raw and normalized spectrograms. Figure on the left is raw signal of quiet pattern with raw spectrogram and normalized spectrogram for patient 3. Figure on the right is raw signal of irritation pattern with raw spectrogram and normalized spectrogram for patient 3. For each plot, power in different frequency bands (y-axis) is shown for each time window (x-axis). The normalization step leads to changes in intensity across different frequency bands, reducing the effect of constant patterns that are not informative, such as the high-intensity 0-30Hz band.

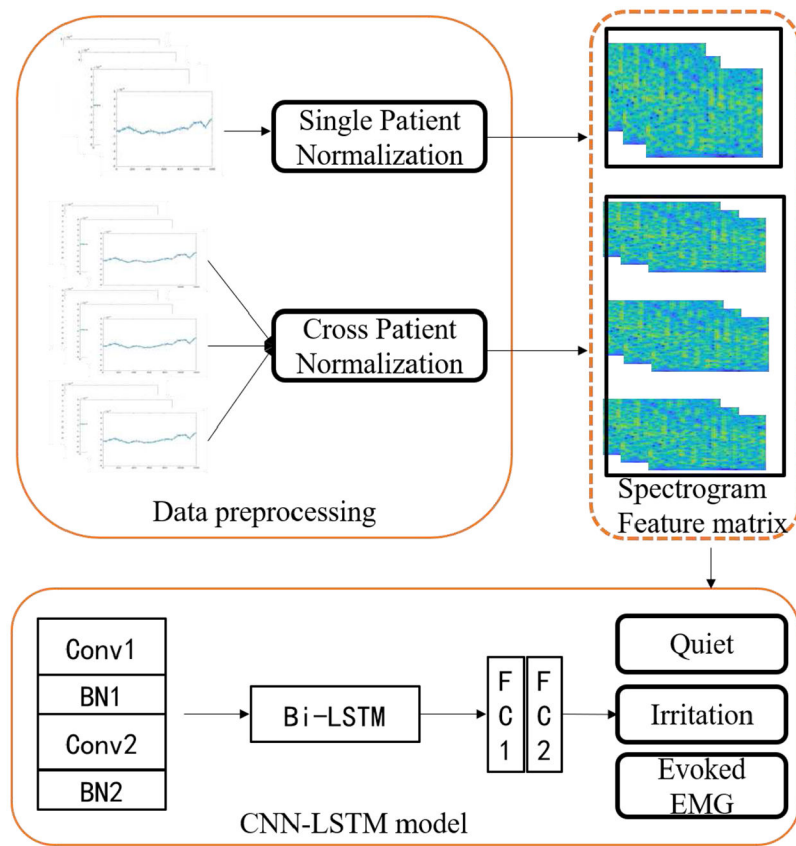


Fig. 5: Analysis pipeline and CNN-LSTM model. After the data preprocessing step, which is different in the single-patient and cross-patient cases, a spectrogram matrix as feature vector is put into the CNN-LSTM model. The CNN-LSTM model is constructed of two convolutional layers and a bidirectional LSTM layer, which is good for capturing both time and frequency features of the EMG signal.

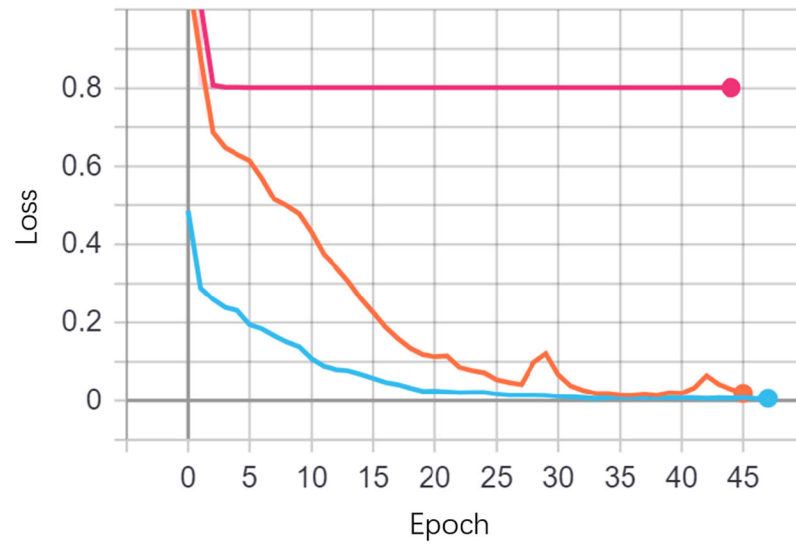


Fig. 6:

Loss curve for cross-patient model. The orange line represents the normal index method, the blue line represents the simplified index method, and the pink line represents the normal index without data normalization. The simplified index method converges more quickly than the normal index method. Without data normalization the method does not converge.

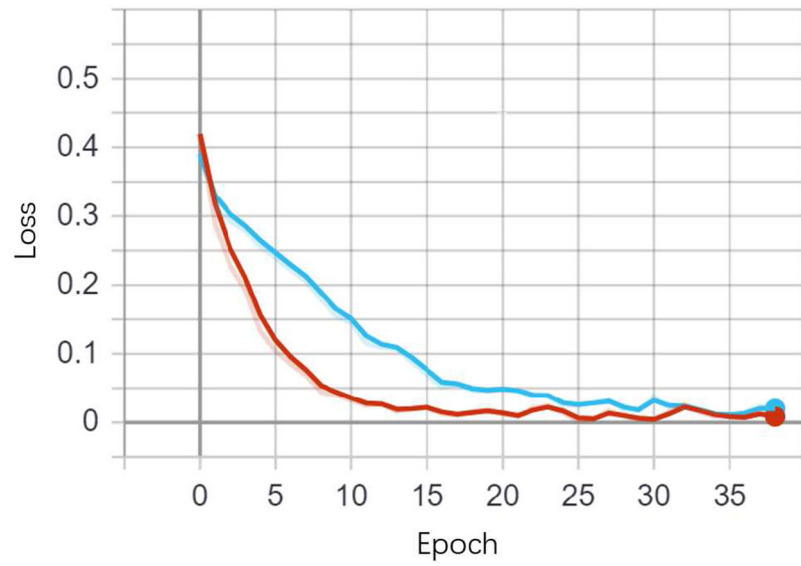


Fig. 7: Loss curve for single-patient model. In these plots, the red line represents normalized data and the blue line represents raw data. In single-patient training, data normalization can help to make training converge faster.

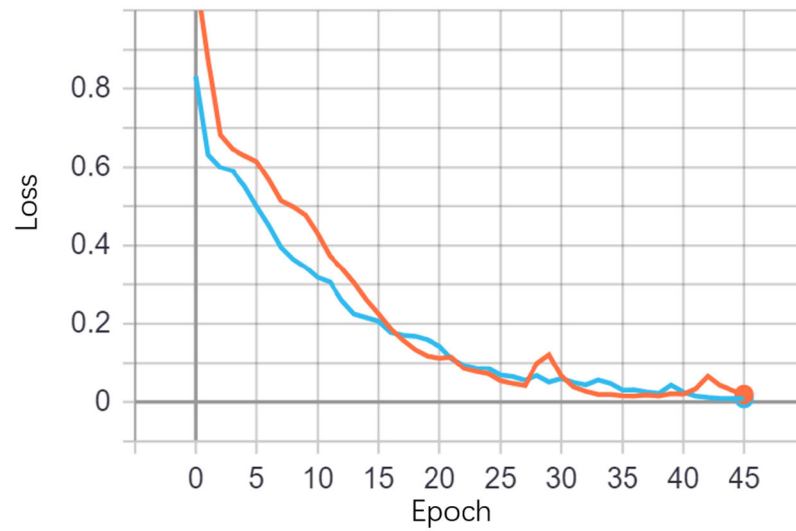
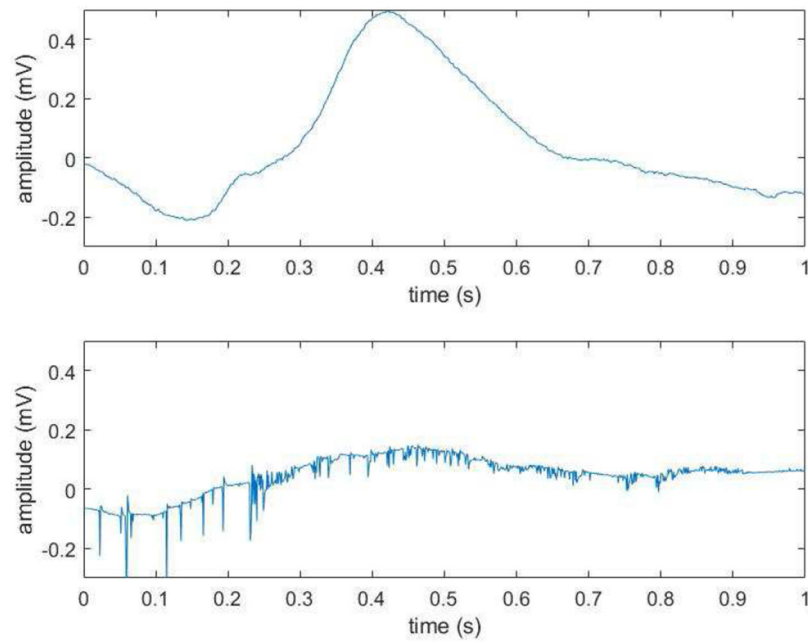


Fig. 8: Loss-curve for cross-patient model. The orange line represents raw data and the blue line represents artifact-free data. The loss curves in these two conditions are close. The model performs well in both cases.

**Fig. 9:**

Examples of erroneous classification. Upper image is Epoch 3917 from patient 1, labeled as weak irritation but classified as quiet. Image below Epoch 979 from patient 1, labeled as artifact but classified as irritation.

TABLE I:

Classes OF EMG Signal Content

Event Class	Activity Name	Description	EMG Signal Waveform
1	Quiet	no activity - intact nerve or possible a sharply transected nerve	
2	Evoked	synchronized with stimulus - identifies existence of nerve - helps locate.	
3	Irritation	sounds like a fluttering sound - indicates working near the nerve	
4	Burst	indication of working near nerve and irritating the nerve in a strong fashion	
5	Injury	continuous train of high-frequency pulses - indicates permanent injury	
6	Artifact	related to electrode manipulation, static from instruments.	

TABLE II:

CNN-LSTM Model Parameters

Layer	Name	Modules	Configurations
1	Conv1	2D convolution	Kernel size: 3*1 Padding: 1*0 Stride: 1
2	BN1	Batch normalization	Feature channels: 4
3	Conv2	2D convolution	Kernel size: 3*1 Padding: 1*0 Stride: 1
4	BN2	Batch normalization	Feature channels: 8
5	Bi-LSTM	Bidirectional LSTM	Hidden units: 256 Dropout probability: 0.2 Recurrent layers: 2
6	FC1	Fully Connected + ReLU	Input size: 512 Output size: 128
7	FC2	Fully Connected	Input size: 128 Output size: class types

TABLE III:

Training AND Testing Data Information

	Class Types	Train Size	Test Size
Normal Index	Quiet	8045	1195
	Regular Irritation	697	83
	Evoked EMG	24	0
	Baseline Irritation	597	0
	Strong Irritation	143	102
	Weak Irritation	13788	3138
	Artifacts	3003	207
Simplified Index	Non-irritation	11072	1402
	Irritation	15225	3323

TABLE IV:

STFT Window Size Analysis

Window Size	Patient 1	Patient 2	Patient 3	Cross patients
50	94.13%	77.65%	82.12%	91.15%
100	96.32%	84.46%	81.46%	94.23%
150	90.25%	77.61%	72.37%	92.34%

TABLE V:

Performance OF Different Models

	Evaluation	Patient 1	Patient 2	Patient 3
GNB	Acc	72.32% $\pm$ 3.0%	80.11% $\pm$ 2.7%	71.57% $\pm$ 2.5%
	Sen	75.56% $\pm$ 3.8%	77.32% $\pm$ 3.4%	73.22% $\pm$ 2.7%
CNN	Acc	88.21% $\pm$ 2.2%	90.15% $\pm$ 1.99%	78.52% $\pm$ 2.2%
	Sen	84.36% $\pm$ 3.3%	95.33% $\pm$ 1.7%	80.49% $\pm$ 2.4%
RNN	Acc	45.74% $\pm$ 3.3%	90.01% $\pm$ 2.0%	84.45% $\pm$ 2.0%
	Sen	47.33% $\pm$ 4.5%	93.27% $\pm$ 2.0%	87.87% $\pm$ 2.0%
CNN-LSTM	Acc	90.31% $\pm$ 2.0%	96.70% $\pm$ 1.2%	86.77% $\pm$ 1.8%
	Sen	87.86% $\pm$ 2.9%	96.32% $\pm$ 1.5%	90.21% $\pm$ 1.8%

Acc represents accuracy which means proportion of correctly identified samples out of all samples.

Sen represents sensitivity of irritation which means proportion of correctly identified irritation samples out of all irritation samples.

TABLE VI:

Model Performance OF Different Index Methods

	Evaluation	Normal Index	Simplified Index
Train	Acc	99.71% $\pm$ 0.1%	99.79% $\pm$ 0.1%
	Sen	98.84% $\pm$ 0.2%	99.83% $\pm$ 0.1%
Test	Acc	89.54% $\pm$ 0.9%	93.14% $\pm$ 0.7%
	Sen	94.23% $\pm$ 0.8%	99.49% $\pm$ 0.2%

TABLE VII:

Performance With/Without Artifacts

	Evaluation	With artifact	Without artifact
Train	Acc	99.71% $\pm$ 0.1%	99.83% $\pm$ 0.1%
	Sen	98.84% $\pm$ 0.2%	98.55% $\pm$ 0.2%
Test	Acc	89.54% $\pm$ 0.9%	89.24% $\pm$ 0.9%
	Sen	94.23% $\pm$ 0.8%	94.39% $\pm$ 0.8%